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Chapter 1: Introduction to Cyber-Forensic OS

The monitor screen flickered. In the dimly lit room, a system alarm sounded softly but steadily. There were intruders on the network. They hadn't hacked in through the front door, but had infiltrated the computer's memory, disguised as ordinary programs. Confidential data was being slowly siphoned out. You don't have much time. Please log in, Analyst.

1.1 Background & Role of Analyst

Welcome to Cyber-Forensic OS. This game is a simulation where you play as a digital detective (Cyber Forensic Analyst) assigned to handle an emergency.

Imagine computer memory (RAM) as a large, busy city. In this city, there are many legitimate workers (legitimate operating system processes) keeping the city running. However, a group of criminals (malware or viruses) has infiltrated the city. They are very clever because they disguise themselves as legitimate workers, so at first glance they appear to be normal programs (e.g. ,`svchost.exe` or `explorer.exe`).

Your job as an analyst is to monitor the city's activities in real time, identifying the real workers and the criminals in disguise. You'll be equipped with tracking devices and a special command terminal to isolate and eliminate these intruders before they steal all your city's confidential data.

1.2 Main Goal of the Game (Objective)

Your mission in this game is very clear: Clear the system of all threats. However, you must race against time and must not act rashly.

- ⏳ Victory Condition (Case Closed): You win the game if you successfully find and eliminate (Execute/Kill) all viruses or malicious processes active in the system. When the screen is clear of threats and the number of malicious processes reaches zero, the system is declared safe.
- 💀 Defeat Condition (Game Over): You will fail this mission if either of the following two things happen:
    1. ***Data Theft Completed (100% Exfiltration):***On your screen, there's a time limit indicator called Data Exfiltrated. If you act too slowly, the

attacker will have successfully removed 100% of your sensitive data, and you'll lose.

2. ***Wrong Target (BSOD / System Crash):***Eliminating viruses shouldn't be done haphazardly. If you misidentify a virus and "kill" the legitimate workers responsible for maintaining the computer (core system processes), the computer will immediately be severely damaged (experiencing a Blue Screen of Death) and the game will be over instantly!

❓ *FAQ Chapter 1 (Frequently Asked Questions)*

Q: Do I need to understand any programming language or real coding to play this game?A: Not at all! Although it looks like a professional hacker tool, this game is designed to be playable by anyone. All the commands you need to type are very simple (such as`help`or`kill [number]`), and there are lots of visual clues (colors) to guide you.

Q: What happens if I just stay quiet and don't type anything?A: The data theft indicator (the yellow Data Exfiltrated bar) will gradually increase. If it reaches 100% for too long, the computer will be taken over by the criminals, and you will lose the game (Game Over).

Q: Then why don't I just delete (kill) all the programs on the screen so it finishes quickly?A: Be careful! Some of the programs on the screen are good programs that keep the computer running. If you delete these good programs, your computer will crash (BSOD) and you'll be dead in seconds. You need to be absolutely sure before "pulling the trigger."

Chapter 2: Anatomy of a Dashboard (User Interface)

A mechanic can't repair an engine if he doesn't know the names and functions of the tools in his toolbox. The same goes for you. The monitor in front of you is your main cockpit. Let's examine each instrument one by one so you don't press the wrong button in a crisis.

The Cyber-Forensic OS interface is designed to be very user-friendly. You don't need to switch between pages or tabs. All the information and tools you need to rescue your computer are available on a single screen.

2.1 Process Tree: Process Hierarchy Map

Located at the top of the screen, this is your main map. The Process Tree displays all the programs currently running on your computer, complete with connecting lines indicating which programs are running (parent and child). Pay close attention to the color of the boxes:

- ***Blue Box (Safe):***This is the core program that protects the operating system (example:`System`,`csrss.exe`). Never touch or delete this program! If you delete it, your computer will immediately crash (BSOD).
- ***Gray Box (Suspicious):***This program should be watched out for because it could be a legitimate program, or a virus in disguise.
- ***Red Box (Evil/Malware):***This is a confirmed criminal. If you see a red box, be prepared to take immediate action.

2.2 Hex Viewer: Peek into Memory Contents

In the bottom left corner, you'll see a complex array of numbers and letters (hexadecimal). Don't panic! This is the Hex Viewer. When you click on any box in the Process Tree, this panel will immediately display the "heads," or raw data, of the program. You don't have to read it all. Simply look for suspicious text patterns on the far right side (such as file names or IP addresses) to help you determine if the program is a virus.

2.3 Forensic Terminal: Command Center

The black panel in the bottom center is where you communicate with the security system. This is your primary weapon.

- ***Information Column:***This screen will always print a flashing red text (`*** ALERT: NEW INFECTION DETECTED! ***`) if a new virus is born.
- ***Type Column (volatility_core >):***This is where you type commands (like hackers in the movies). For example, typing`kill`to destroy the program, or`help`to see a list of commands you can use.

2.4 Properties & Metadata: Reading Program ID Cards

Located in the upper right corner. When you click a box in the Process Tree, this panel will display the program's "ID," or identity. The most important information to note here is the PID (Process ID). This is the program's unique identifier, and it's what you'll type into Terminal if you want to delete it.

2.5 Active Threat Monitor: Countdown to the Data Apocalypse

Located just below Metadata, this yellow bar indicates your time limit (Data Exfiltrated). Pay close attention! The more red boxes (malware) on the screen, the faster this yellow bar fills. If this bar fills to 100%, the hacker has successfully stolen all your confidential data, and you're out.

2.6 Live Threat Spectrum: Computer Health Graph

Located in the bottom right corner, this is an instant monitoring tool. This bar graph will show your computer's health status. The longer the blue line, the safer your computer is. However, if the graph is predominantly bright red, the infection has spread and you need to act immediately before it's too late.

❓ *FAQ Chapter 2 (Frequently Asked Questions)*

Q: What is the function of the connecting red lines in the Process Tree?A: The red line indicates that the virus (parent) is trying to spread the infection and give birth to new viruses (daughters). This is a clue for you to find who the main culprit is!

Q: Why are there programs with the exact same name (for example, two powershell.exe), but different colors?A: That's how cunning viruses are! They often use the names of legitimate Windows programs (like PowerShell) to avoid

suspicion. You can't just rely on the program name; you need to scan it first or track its PID number.

Q: I'm having trouble reading Hex Viewer because there are too many numbers. Is this a must-read?A: Not required. Hex Viewer is just an additional tool. If you're feeling overwhelmed, you can completely rely on Terminal and Scanner Tools, which will be discussed in the next chapter.

Chapter 3: Threat Dynamics (Gameplay Mechanics)

Don't be fooled by the calm appearance of the screen. Inside the memory chamber, a war is raging silently. Computer viruses aren't inanimate objects; they're intelligent entities constantly multiplying, infiltrating, and seeking vulnerabilities. If you simply sit back and watch, they'll soon take over the entire city. Know your enemy, understand how they operate, and cut to the root of the problem.

Understanding how threats operate in Cyber-Forensic OS is key to winning the game without feeling overwhelmed. Let's dissect the types of enemies you'll face.

3.1 Threat Level Classification

Each program (box) that appears in the Process Tree has its own level of threat. You can easily distinguish them by the color they appear in:

- ***Safe (Blue):***These are civilians or legitimate city guards (such as`System`or`smss.exe`). They work purely to keep the computer running. It is strictly forbidden to touch them, or the system will crash (BSOD).
- ***Suspicious (Gray/Orange):***This is a program that's behaving strangely. It may have no known parent, or it may be using a fake name. You should scan this gray box first to determine whether it's friend or foe.
- ***Malicious (Red):***This is a genuine criminal. This program actively siphons your data and speeds up exfiltration time. If you see a red box, be prepared to delete it.
- ***Critical:***This is a major criminal. Typically, this type of red box hides a secret address (IP address) within its memory. Finding it will earn you a very lucrative bonus.

3.2 Regular Troops vs. the Mother (Root Infector)

Not all red boxes are created equal. Among the swarm of malware on the screen, they fall into two roles:

1. ***Common Threats (Troops):***They're little creatures that reside in memory and steal data. Killing them is crucial, and each time you destroy a regular troop, the yellow bar (stealing time) will decrease slightly (a 10% bonus).

2. ***Root Infector (The Parent):***This is the Main Boss, the mastermind behind all the chaos, also known as Patient Zero. The Mother is usually the first virus to enter a computer and hide. Finding and killing the Mother is your top priority, as it grants a massive time recovery bonus (25% bonus)!

3.3 How Viruses Spread (Dynamic Process Migration)

This game will become increasingly tense as time goes on. Why? Because the Root Infector won't stay still.

- ***Giving Birth to New Troops:***As long as the Mother Virus is alive on your screen, it will spawn new viruses (child processes) every 15 to 20 seconds.
- ***Snowball Effect:***Each time a new virus is born, the Terminal alarm will sound, and you'll see a new red line connecting the mother to its new offspring. If you ignore it, your screen will soon be filled with red.
- ***Maximum Limit:***Don't worry, this game is still humane. The maximum infection rate is limited to 8 active viruses on screen at a time. However, if there are 8 active viruses at once, the yellow data theft bar will move very quickly!

Key to Victory:Don't panic when you see a bunch of red boxes popping up. Take a deep breath, trace the red lines upward, find the top box (the Mother), and destroy it first! Once the Mother dies, they won't be able to reproduce anymore.

❓ *FAQ Chapter 3 (Frequently Asked Questions)*

Q: Can the Mother disguise herself as blue (Safe)?A: No. Our security system is quite sophisticated. Processes that are truly core and crucial to the operating system are always blue and guaranteed to be safe. The Mothership is usually disguised as gray at first, then turns red. Remember: Never "kill" the blue box!

Q: What if there are 8 viruses on the screen, is the game immediately Game Over?A: It's not an immediate game over. However, with eight viruses sucking up data simultaneously, the Data Exfiltrated bar (your time limit) will shoot up like crazy. You might only have a few dozen seconds before reaching 100%.

Q: Can child viruses (regular troops) turn into new Hosts?A: In this version of Cyber-Forensic OS, the Parent is usually present at the beginning (case initiation).

However, it is highly recommended to always trace the top of the hierarchy (lines) to ensure you are cutting the threat at its root.

Chapter 4: Conducting Analysis & Investigation

Before pulling the trigger, a sniper will observe their target through binoculars. In the world of cyber forensics, guesswork is a big no-no. You shouldn't just execute a program just because its name sounds strange. The "Tools & Analyzers" panel is your mini forensic laboratory. Let's learn how to gather evidence to confirm your target is truly a criminal, while also gaining time in the process.

In this game, analysis tools aren't just for browsing; they provide real tactical advantages. You have two main scanners at the bottom right of the screen: the Regex Engine and the YARA Matcher.

4.1. Regex Engine: Text Search Magnifier

Regex (Regular Expressions) is a smart search tool. Think of it like the "Search" or "Find" feature, but much more powerful.

- ***Its function:***This tool is very suitable to use if you suspect that there is a virus that is trying to contact hackers out there, and you want to find the secret address (IP Address) hidden in the brain of the virus.
- ***How to use:***Click the program you suspect in the Process Tree. Then, type the number pattern (for example:`19.11.2.*`) in the Regex box, and press Enter. The system will immediately rummage through the program's memory for a match.

4.2. YARA Rules: Digital Wanted Posters

If Regex is used to search for specific text, YARA is like matching a person's face to a police wanted list. YARA checks to see if the program's "behavior," or structure, is similar to that of known, real-world viruses.

- ***Its function:***It is very powerful in exposing the disguise of gray boxes (Suspicious) that try to hide behind the name of official Windows programs.
- ***How to use:***Click the suspected program. Then, click the YARA drop-down menu and select a "Rule," such as detecting code injection or searching for encryption keys (typical of ransomware). Once you select one, the tool will automatically run.

4.3. Time Bonus from Evidence Collection (IoC)

Scanning and identifying threats isn't just a waste of time. Every time your scanner (Regex or YARA) successfully finds a match in the red or gray boxes, the discovery is recorded as an IoC (Indicator of Compromise) or "Evidence of Compromise."

- ***Super Benefits:***Every time you find 1 Evidence of Crime (IoC), the system will reward you by reducing the yellow bar of data theft (Exfiltration) by 5%.
- ***Strategy:***If your time limit is very low (for example, the yellow bar has reached 90%), you can deliberately scan viruses that are clearly red to get a time bonus, giving you a "breath of fresh air" before you start executing (killing) them.

🕵 *Example of an Investigation Case Study*

Emergency Situation:Critical condition! Your data theft bar has reached 80%. On the screen, you see a program`svchost.exe`(blue/safe box) gives birth to a new program called`wscript.exe`(grey box). You suspect it's a virus, but you need certainty and also need to buy time to avoid losing too quickly.

Your Detective Steps:

1. ***Target Highlight:*** You click on the grey bo x`wscript.exe` (PID: 31255).
2. ***Quick Scan:*** You don't delete it immediately, but instead run to the Tools & Analyzers panel, then select Rule: Find C2 Beaconing on the YARA menu.
3. ***Identification Results:***The terminal in the bottom center of the screen suddenly flashes red and prints:`>> THREAT DETECTED! IoC Registered.`(EVIDENCE FOUND!)
4. ***Time Prize:***Thanks to the discovery of this evidence, the data theft bar, which was previously critical (80%), immediately dropped to 75%!
5. ***Conclusion:***Now you have a few extra precious seconds, and you are 100% sure that `wscript.exe` It's a malicious malware. It's time to prepare your "weapons" in Terminal to destroy it.

❓ *FAQ Chapter 4 (Frequently Asked Questions)*

Q: Do I have to scan all the red boxes before deleting them?A: Not required. If the box is already flashing red, you're allowed to kill it immediately. Scanning is only

useful for validating gray boxes or if you're specifically aiming for the 5% time reduction bonus.

Q: What happens if I scan the secure blue box?A: Nothing bad will happen. The terminal will just display a green message.`>> Scan complete: Process is clean.`However, you are just wasting your valuable time as there will be no bonuses awarded.

Q: If scanning gives bonus time, why don't I just scan for the same virus over and over again until the time runs out?A: True victory is achieved only when all viruses are eliminated (threat count = 0). Continually scanning for viruses without ever killing them is simply delaying inevitable defeat, as more malware will continue to suck up your data!

Chapter 5: Active Mitigation & Terminal Usage

After observing and gathering evidence, it's time to act. Scanning for viruses will only slow them down, but not stop them. The dark terminal in front of you is your primary execution weapon. This is where you type commands like a hacker or high-level systems engineer. Keep your fingers on the keyboard, ensure your aim is precise, and avoid missing a target.

The Forensic Terminal, located in the bottom-center panel, is your direct communication bridge with the computer's core operating system. Here, you're no longer just a spectator, but in complete control.

5.1 Basic Investigation Orders

While the Process Tree screen is visually appealing, sometimes you need quick text-based information. You can type the following commands in Terminal (right next to the text `volatility_core >`) then press Enter:

- **`Help`** : Emergency command. Type this if you forget what commands you can use. The system will display a list.
- **`clear`**: Clears the Terminal screen. Very useful if the screen is cluttered with red warnings and you need some space to focus.
- **`Pslist`** : Displays a neat text list of all currently running programs (along with their PID numbers). This feature is very helpful if the Process Tree is too cluttered due to viruses.
- **`malfind`**&**`netscan`** : This is an alternative way to perform a scan (same as the YARA and Regex buttons in the previous chapter), but done via terminal typing.

5.2 Threat Execution: Pulling the Trigger with `kill`

Here's how you can get rid of malware and win the game.
Commands`kill`(kill/terminate) is used to forcibly kill a malicious program.

- ***How to Use It:***
    1. Click the red box (virus) in the Process Tree that you want to destroy.
    2. Look at the KTP (Properties & Metadata) panel on the top right, and note down the PID number (e.g. PID:`41965`).

3. Hover over Terminal, then type:`kill 41965`(make sure there is a space after the word kill).
4. Press Enter.

- ***Signs of Success:***If your guess is correct, Terminal will print a green message stating the process has been successfully killed, a success beep will sound, and the yellow data theft bar will decrease (gaining bonus time).

5.3 Fatal Risk: Wrong Execution and Blue Screen of Death (BSOD)

With great power comes great responsibility. Order`kill`no matter who you point it at; it will instantly kill its target.

STRICT WARNING:If you are careless and type a command`kill`on the PID number of the blue box (safe/legitimate program), you have just made the most fatal mistake in the world of cyber forensics.

- The computer will immediately lose its foundation and fall apart (experiencing the Blue Screen of Death or blue screen).
- The terminal will print a critical error warning, the data theft bar will immediately jump to 100%, and you will immediately be declared a LOSS (Game Over).

✵ Terminal Mitigation Case Study Example

Emergency Situation:Your screen is messed up! A red node named`rundll32.exe`A new infection has just appeared and the Terminal is flashing warnings of new infections. Your exfiltration bar is rapidly approaching 80%. You need to stop it immediately.

Execution Steps (Mitigation):

1. ***Target Search (Aiming):***You calmly click on the red box`rundll32.exe`In the top right panel, you read his ID card number: PID: 36495.
2. ***Pull the trigger:***You quickly move the mouse cursor to the typing field in the black Terminal, then quickly type:`kill 36495`and press Enter.
3. ***Results (Target Fallen):***A success notification sound is heard. The terminal prints a bright green message : `SUCCESS: Malicious process "rundll32.exe" (PID: 36495) terminated.`

4. Box `rundll32.exe` The virus will shut down and stop sucking up your data. The yellow bar will slow down, and you can breathe a sigh of relief while hunting down the remaining viruses.

❓ *FAQ Chapter 5 (Frequently Asked Questions)*

Q: I am typing `kill powershell.exe` ***but it says "Usage error". What's wrong?***J: Command `kill` does not recognize program names (because names can be faked). It only recognizes identification numbers. You must use the PID number. The correct format is `kill [space] [PID Number]`, example: `kill 1234`.

Q: I have typed `kill 8948`***, but it says "Process not found". Why?***A: There are two possibilities: First, you mistyped the PID number. Second, you may have already killed the virus with that number. Always make sure the PID number you type matches the one in the active box on the screen.

Q: Why do I need an order? `Pslist` ***if I can see everything in the Process Tree image?***A: Early in the game, viewing the Process Tree is easy. However, as the infection spreads further, the tree can become very dense.`pslist`will give you a neat list of numbers and names lined up straight from top to bottom, making it easy for you to find the PID number without being distracted by the image.

Chapter 6: End Game Conditions (Win/Loss)

Every crisis operation has an ending. In the world of cybersecurity, there's no such thing as a "tie" or a "balance." You have only two possible outcomes at the end of this mission: returning home a hero who saved all the company's secrets, or watching your screen turn red as hackers laugh at the destruction of your system. Let's clarify what determines the outcome of this game.

Just like in a competition, you must know exactly how to score a win and what will get you disqualified. In Cyber-Forensic OS, your ultimate goal is absolute.

6.1 Victory Conditions: Completely Complete Without Leftovers

The only way to win this game is to completely clean the computer. You must eliminate all active virus boxes (both the parent and its children) until the threat count reaches ZERO.

- ***Execution is Key:***Please always remember, simply scanning for viruses and collecting evidence of crime (IoC) will not win you the victory. Evidence of crime only gives you more time (reducing the yellow bar percentage). True victory only comes from the command line.`kill`.
- ***Victory Moment:***If you successfully type the command`kill`At the last virus remaining on the screen, the game will immediately stop. The terminal will print a proud green message: `SUCCESS: ALL THREATS CONTAINED. YOU WON!`. The time bar will turn bright blue, and the screen will display "CASE SOLVED." Congratulations, you've succeeded!

6.2 Defeat Conditions: Too Slow or Too Careless

Unfortunately, it's very easy to fail this mission. You'll lose (Game Over) if either of these two nightmares occurs:

1. ***Data Theft Completed (100% Exfiltration) - Defeat due to slowness***If you panic too long, scan too long, or fail to find the Mother, the yellow "DATA EXFILTRATED" bar will continue to creep up relentlessly. When this bar reaches 100%, it means the hacker has completely removed your sensitive data. Your computer has been completely compromised. The terminal will

beep loudly and display a red warning:`CRITICAL: EXFILTRATION COMPLETE. SYSTEM COMPROMISED.`

2. ***Fatal Error / Blue Screen (BSOD) - Defeat due to carelessness***As discussed in the previous chapter, deleting (killing) the core security system program (the blue box) is the greatest sin. If you type the wrong PID number and kill the system guard instead, the computer will immediately crash. The timer will immediately jump to 100%, and you will be declared defeated with a message. `CRITICAL ERROR... System crashing...`. In this game, being careless is much more dangerous than being slow.

❓ *FAQ Chapter 6 (Frequently Asked Questions)*

Q: My yellow bar is already at 99%, and I managed to kill the last virus. Did I win or lose?A: You WIN! As long as the yellow bar hasn't actually reached 100%, the system can still be saved. These moments in the 90s are what usually make the game feel really tense.

Q: I have successfully scanned 3 viruses and found the evidence, why haven't I won?A: Because scanning is not a way to win. Scanning only reduces the yellow bar (giving you an extra 5% time). To win, you must use the command`kill`in Terminal to destroy the viruses.

Q: Is there a pause button if I want to go to the toilet?A: This is a real-time hacking emergency simulation. Just like a real-world hacking incident, the hacker won't wait for you to go to the bathroom. Time will keep ticking! Make sure you're fully focused before starting a New Case.

Chapter 7: Advanced Strategies & Tips

A novice will shoot anything that moves in panic. An expert will take a deep breath, read the attack pattern, and take down the enemy leader with a single, well-aimed bullet. You have now mastered all the tools in this control room. In this final chapter, we will transform you from a mere analyst into an unstoppable cyber defense commander. Here's the secret to conquering the worst-case scenario.

Typing quickly helps, but a calm strategy is far more decisive for victory. Here are some tactics veteran players use to control the situation when the system is on the verge of collapse.

7.1. Cut the Head of the Snake: Prioritize Root Infectors

The most common mistake new players make is to focus on eliminating the child viruses (the newly spawned red boxes) one by one. Remember, as long as the mother (Root Infector) is alive, it will continue to spawn new offspring every 15 to 20 seconds. Killing the offspring is like mopping the floor while the faucet is still leaking.

- ***Tactics:***When you see a branching red line in the Process Tree, don't immediately shoot at the bottom end. Follow the red line upwards towards the center of the branch. The box that spawned the branches is the Mother! PNG
- ***Profit:***Killing a normal croco only lowers the exfiltration bar by 10%. However, executing the Mother will permanently stop the birth of new viruses and grant a bonus 25% reduction in the time bar!
- ***Sample case:***On the screen appears `powershell.exe` (PID: 41843) and`c md.exe` (PID: 41965) which is lit red. The red lines from these two boxes point to the one box above them, namely `svchost.exe` (PID: 8948). Ignore `powershell` And`cmd`for the time being. Just type **`kill 8948`**At the Terminal! After the mother is down, you can then leisurely clean up the remains of her offspring.

7.2. Rapid Triage: Controlling Panic When the Screen Goes Red

When alarms are blaring, red warning text fills the Terminal, and the Live Threat Spectrum graph is dominated by red, panic is your worst enemy. Panic makes your

fingers rush and risks mistyping the PID of the malicious program (triggering a BSOD).

- ***Tactics:*** Calm down. The system has a maximum limit mechanism where only eight viruses can be active at a time. Your screen will never become so overrun with hundreds of viruses that it's impossible to control.
- ***Sample case:*** The exfiltration bar has reached 85% (very critical) and there are 6 red boxes on the screen. Don't panic when typing . `kill` Randomly! Use 5 seconds to trace a line towards the Mother, then kill her first. The time bar will immediately drop to 60%, giving you plenty of breathing room to meticulously execute the remaining 5 viruses one by one.
  PNG

7.3. When to Scan vs. When to Kill

Scanner tools (YARA, Regex,`malfind`,`netscan`) is very useful, but it requires clicks and time. You have to know when the right time to use it:
PAS

- ***When to Scan:***
  1. When you are faced with a gray box (Suspicious) and want to confirm whether it holds evidence of a crime or not.

  2. When the time bar is at the end of its rope (say 90%) and you need an instant "breathing" of 5% before you can look up the Master PID number.
- ***When to Execute (Kill) Immediately:***If a box is clearly lit up Bright Red (Malicious) in the Process Tree (such as `wscript.exe` or `rundll32.exe`), don't waste time re-scanning! Scanning it only reduces the time bar by 5%, while typing it directly `kill <PID>` will reduce the time bar by 10% (or 25% if it's Parent) while permanently slowing the rate of data theft.

❓ *FAQ Chapter 7 (Frequently Asked Questions)*

Q: What is the fastest way to differentiate **`svchost.exe`** ***the original (safe) with*** **`svchost.exe`** ***fake (virus)?***A: Look at the color and see the lineage! The original and vital system programs for the computer will always be Blue. If there are any

`svchost.exe` is Gray or Red, or it suddenly gives birth to a program like `cmd.exe` with the red line, it can be 100% certain that it is an intruder in disguise!

Q: I've killed the Mother, why does the yellow bar (Data Exfiltrated) keep going up?A: Because the previously spawned virus offspring are still alive! Even though no new viruses are being spawned, the remaining active troops on screen will continue to suck up your data. You must wipe them all out until the threat count reaches 0.

Q: If I have won or lost, how do I start a new game quickly?A: You don't need to close the application. Just type the command `restart`or`new case` in Terminal, then press Enter. The system will immediately reset memory and randomize a new attack scenario for you to solve!

Walkthrough Simulation: "Silent Siphon" Cleaning Operation

It was a peaceful night in the cybersecurity control center, until the main monitor suddenly lit up. CYBER-FORENSIC OS v2.4 automatically initialized and loaded a memory dump from the server computer suspected of being compromised. Time passed, and the data exfiltration rate began to rise from 0%. Here's a step-by-step look at how an analyst solved this case from start to finish.

Stage 1: Initial Observation & Terrain Mapping (Seconds 0 to 15)

As soon as the game starts, the Terminal displays an opening message that a new case has begun. You immediately glance at the Process Tree panel at the top and the Live Threat Spectrum in the bottom right corner.

1. ***Reading a Spectrum Graph:***In the bottom right corner, you can see the initial status: `Safe: 3`, `Suspect: 1`, `Malware: 2`. This means there are a total of 3 initial threats hiding among 3 legitimate system programs.
2. ***Observing the Process Tree:***

- There are three Blue (Safe) boxes: `System` (PID: 4), `smss.exe` (PID: 368), and `csrss.exe` (PID: 544). You mark all three in your head as forbidden areas—never touch those PIDs to avoid triggering a Blue Screen of Death (BSOD).

- There is one Gray box (Suspect): `explorer.exe` (PID: 8552).
  PNG
- There are two Red boxes (Malware): `svchost.exe` (PID: 8948) and `lsass.exe` (PID: 7123).
  PNG

Stage 2: Malware Counterattack & Parent Identification (18th second)

While you are checking the existing boxes, suddenly a warning sound is heard from the system!
PAS

- ***Spread of Infection:***The Terminal screen flashes red displaying the message:`*** ALERT: NEW INFECTION DETECTED! ***`

  `Suspicious process spawned: wscript.exe (PID: 31255)`

- ***Tracking the Red Line:***In the Process Tree, a new red box appears named `wscript.exe` (PID: 31255). Most importantly, there is a bright red connecting line extending from `svchost.exe`(PID: 8948) go straight to `wscript.exe` the!

- ***Tactical Conclusion:***The red line is absolute proof.`svchost.exe` (PID: 8948) is no ordinary virus, but a Root Infector (the parent) that spawns new malicious processes! The total number of active threats has now increased to 4, and the yellow Data Exfiltrated bar is starting to climb to 15%.

Stage 3: Investigation & Buying Time (25th Second)

Before starting to execute, you want to check the grey box. `explorer.exe` (PID: 8552) to confirm its status while seeking a bonus reduction in exfiltration time.
PNG

1. ***Click Target:***You click on the grey box `explorer.exe` in the Process Tree. The box now has a white border. The Properties & Metadata panel at the top right displays:`PID: 8552`,`Process: explorer.exe`,`Parent: Unknown (Orphaned)`. In the bottom left Hex Viewer panel, you can see a series of random hexadecimal codes with a suspicious pattern.`EB 10 90 90 E8`red in the middle.

2. ***Running a Scan:***You hover over the Tools & Analyzers panel on the right, click the Select YARA Rule... dropdown menu, and then choose**`Rule: Detect Reflective Injection`**.

3. ***Scan Results:***The terminal prints: `Running YARA scan on PID 8552...`
   `>> THREAT DETECTED! IoC Registered.`

   Proven! The gray box was malware in disguise. As a reward for the evidence discovery (IoC), the Data Exfiltrated bar, which had reached 18%, immediately dropped 5% to 13%.
   PAS

Stage 4: Cutting Off the Snake's Head / Mother Execution (35th Second)

Before `svchost.exe` (PID: 8948) gave birth to a second virus child, you must kill it immediately.

1. You click on the type field in the Forensic Terminal (`volatility_core >`).
2. You type the command:**`kill 8948`**then press Enter.

3. ***Execution Results:***
   - A beep sound is heard indicating success.
   - The terminal prints bright green text:`SUCCESS: Malicious process "svchost.exe" (PID: 8948) terminated.`

   - Since you just killed a Root Infector, you gain a 25% time recovery bonus! The Exfiltrated Data bar immediately drops back to 0%!
   - More importantly, the chain of spread had been broken. There will be no more new viruses appearing suddenly.

Stage 5: Clean Sweep of Remaining Troops (45th to 75th Seconds)

Even though the parent is dead, there are still three malicious processes remaining in memory (two original processes and one previously spawned child). To avoid confusion when viewing the tree, you can use the text list command in Terminal:
PAS

1. ***Viewing Active List:***You type `pslist` in Terminal and press Enter. Terminal displays a neat list of all running processes along with their PID numbers.
2. ***Second Target Execution (Virus Child):***You type`kill 31255`(to turn off `wscript.exe`) then press Enter. Terminal: `SUCCESS: Malicious process "wscript.exe" (PID: 31255) terminated.`(Remaining threats: 2).
3. ***Third Target Execution (Static Malware):***You see a red box`lsass.exe` on the screen has PID `7123`. You type `kill 7123` then press Enter. Terminal: `SUCCESS: Malicious process "lsass.exe" (PID: 7123) terminated.` (Remaining threats: 1).

4. ***Final Target Execution (Undercover Suspect):***Lastly, there is just a gray box `explorer.exe` (PID: 8552) which we have previously proven to be malicious through YARA Scan. You type `kill 8552` and press Enter. Terminal:
   `SUCCESS: Malicious process "explorer.exe" (PID: 8552) terminated.`

Stage 6: Victory (Case Solved!)

As per the order `kill 8552` executed, the number of active threats in the official system touched 0.

- ***Visual Changes:***The Live Threat Spectrum graph in the bottom right corner has now changed to 100% Blue (Safe: 3, Suspect: 0, Malware: 0, Critical: 0).
- ***System Status:***The text in the Active Threat Monitor panel changes to `CASE SOLVED - THREAT CONTAINED` with the indicator bar changing color to a calm golden blue.
- ***Closing Message:***The game clock stops, and the Terminal prints the winning message : `SUCCESS: ALL THREATS CONTAINED. YOU WON!`
  `Type "newcase" for another challenge.`

The operation completed in less than a minute and a half without a single system process being harmed!